

Computer Networking

BCA — Semester 5 — 2020 — Answer Sheet

Subject Code: CACS303

Group B

Attempt any SIX questions. [6×5=30]

2. Define protocols. Explain WWW and HTTP protocol.

[1+2+2]

Answer: Protocol: A protocol is a set of rules and conventions that govern how data is transmitted, received, and processed in a network. Protocols define the format, timing, sequencing, and error checking for communication between devices. They ensure that different systems can communicate effectively despite differences in hardware, software, or architecture. **Characteristics of Protocols:** • **Syntax:** Structure or format of data (how it is organized) • **Semantics:** Meaning of each section and action to take • **Timing:** Speed matching and sequencing of data **WWW (World Wide Web):** The WWW is an information system where documents and other web resources are identified by URLs, interlinked by hyperlinks, and accessible via the Internet. It was invented by Tim Berners-Lee in 1989. **Key Components of WWW:** • **Web Browser:** Client software (Chrome, Firefox) that requests and displays web pages • **Web Server:** Software (Apache, Nginx, IIS) that stores and serves web content • **URL (Uniform Resource Locator):** Address that identifies a resource on the web • **HTML:** Markup language for creating web pages • **HTTP/HTTPS:** Protocol for transferring web pages **HTTP (Hypertext Transfer Protocol):** HTTP is an application-layer protocol used for transmitting hypermedia documents, such as HTML. It is the foundation of data communication on the WWW. **Features of HTTP:** • **Stateless:** Each request is independent; server does not retain client information between requests • **Connectionless:** Client and server connect only during request/response • **Uses TCP port 80 (HTTP) or 443 (HTTPS)** • **Supports various methods:** GET, POST, PUT, DELETE, HEAD, OPTIONS **HTTP Request Methods:** • **GET:** Requests data from server (retrieval) • **POST:** Submits data to server (form submission) • **PUT:** Updates existing resource • **DELETE:** Removes specified resource • **HEAD:** Same as GET but returns only headers **HTTP Response Codes:** • **1xx:** Informational (100 Continue) • **2xx:** Success (200 OK, 201 Created) • **3xx:** Redirection (301 Moved Permanently) • **4xx:** Client Error (404 Not Found, 400 Bad Request) • **5xx:** Server Error (500 Internal Server Error)

3. Define transmission impairment. Explain the causes of impairments.

[1+4]

Answer: Transmission Impairment: Transmission impairment refers to the degradation of signal quality during data transmission over a communication channel. As signals travel through media (cable, fiber, air), they become weaker, distorted, or contaminated by noise, leading to data errors. **Causes of Transmission Impairment:** 1. **Attenuation:** - Loss of signal energy as it travels through the medium - The signal strength decreases with distance - Measured in decibels (dB) - **Solution:** Use amplifiers (analog) or repeaters (digital) to boost the signal - Example: A voice signal over a long telephone line becomes weaker and harder to hear 2. **Distortion:** - Change in the shape or form of the signal - Occurs because different frequency components travel at different speeds in the medium - Results from non-linear characteristics of the transmission channel - In composite signals (made of multiple frequencies), each frequency component may arrive at different times - **Solution:** Use equalizers to correct frequency response - Example: Audio distortion in poor quality speakers 3. **Noise:** - Unwanted signals inserted between transmitter and receiver - Types of noise: • **Thermal Noise:** Random motion of electrons in a conductor, present in all electronic devices. Cannot be eliminated. • **Induced Noise:** Caused by sources like motors and appliances. Can be minimized with shielding. • **Crosstalk:** Unwanted coupling between adjacent wires. One wire acts as antenna transmitting, another as antenna receiving. • **Impulse Noise:** Sudden spikes caused by external events like lightning, power surges. Brief but high amplitude. • **Shot Noise:** Random fluctuations in current flow in electronic devices. - **Solution:** Shielding, filtering, proper grounding, using better cables **Signal-to-Noise Ratio (SNR):** $SNR = \text{Signal Power} / \text{Noise Power}$ Higher SNR means better signal quality. Measured in decibels. **Impact on Digital vs Analog:** • Digital signals can be regenerated by repeaters • Analog signals can only be amplified (noise is amplified too) • Digital transmission is more resistant to noise

4. Define HDLC. Explain the HDLC frame formats.

[1+4]

Answer: HDLC (High-Level Data Link Control): HDLC is a bit-oriented synchronous data link layer protocol developed by ISO. It is widely used for reliable point-to-point and multipoint communication. HDLC provides error detection, flow control, and supports various operational modes. **Features of HDLC:** • Bit-oriented (uses bit stuffing for transparency) • Synchronous transmission • Supports full-duplex communication • Provides error detection using CRC • Supports both point-to-point and multipoint configurations **HDLC Frame Format:** | Flag | Address | Control | Information | FCS | Flag | | 8 bits | 8 bits | 8/16 bits | Variable | 16/32 bits | 8 bits | 1. **Flag Field (8 bits):** - Pattern: 01111110 - Marks beginning and end of frame - Used for synchronization 2. **Address Field (8 bits):** - Contains address of secondary station - In point-to-point: usually 11111111 (broadcast) or specific address - In multipoint: identifies destination or source station - First bit indicates if address is extended (1 = extended) 3. **Control Field (8 or 16 bits):** Defines the type and function of the frame. Three types: a) **I-Frame (Information Frame):** - First bit = 0 - Carries user data - Contains send sequence number N(S) and receive sequence number N(R) - Used for piggybacked acknowledgments b) **S-Frame (Supervisory Frame):** - First two bits = 10 - Used for flow control and error control (no data) - Types: RR (Receive Ready), RNR (Receive Not Ready), REJ (Reject), SREJ (Selective Reject) c) **U-Frame (Unnumbered Frame):** - First two bits = 11 - Used for link management (setup, disconnection, mode setting) - Examples: SNRM (Set Normal Response Mode), SABM (Set Asynchronous Balanced Mode), DISC (Disconnect) 4. **Information Field (Variable):** - Contains actual data payload - Present only in I-frames and some U-frames - Variable length 5. **FCS (Frame Check Sequence) - 16 or 32 bits:** - Contains CRC (Cyclic Redundancy Check) for error detection - Computed over Address, Control, and Information fields - Receiver recalculates and compares **HDLC Operational Modes:** • **NRM (Normal Response Mode):** Unbalanced - secondary responds only when polled by primary • **ABM (Asynchronous Balanced Mode):** Balanced - both stations are equal (most common) • **ARM (Asynchronous Response Mode):** Unbalanced - secondary can initiate without polling

5. Define IP Address. Specify IPv4 address classes with their address ranges.

[1+4]

Answer: IP Address: An IP (Internet Protocol) address is a unique numerical identifier assigned to each device connected to a computer network that uses the Internet Protocol for communication. It serves two main functions: host/network interface identification and location addressing. **Characteristics:** • IPv4: 32-bit address written as four octets (e.g., 192.168.1.1) • IPv6: 128-bit address written in hexadecimal (e.g., 2001:0db8:85a3::8a2e:0370:7334) • Each IP address consists of a network portion and a host portion • Assigned by IANA and regional registries **IPv4 Address Classes:** ClassFirst BitsNetwork BitsHost BitsAddress RangeDefault MaskUse A08241.0.0.0 to 126.255.255.255255.0.0.0 (/8)Large networks B101616128.0.0.0 to 191.255.255.255255.255.0.0 (/16)Medium networks C110248192.0.0.0 to 223.255.255.255255.255.0 (/24)Small networks D1110--224.0.0.0 to 239.255.255.255-Multicast E1111--240.0.0.0 to 255.255.255.255-Reserved/Experimental **Special IP Addresses:** • **127.0.0.0/8:** Loopback addresses (127.0.0.1 = localhost) • **0.0.0.0:** Default route / unspecified address • **255.255.255.255:** Limited broadcast • **10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16:** Private IP ranges (RFC 1918) • **169.254.0.0/16:** Link-local addresses (APIPA) **Class A Details:** • Supports 126 networks (0 and 127 reserved) • Each network supports 16,777,214 hosts • Used by very large organizations (government, ISPs) **Class B Details:** • Supports 16,384 networks • Each network supports 65,534 hosts • Used by medium to large organizations **Class C Details:** • Supports 2,097,152 networks • Each network supports 254 hosts • Most common for small businesses and home networks

6. Define Subnetting. Suppose you are given network address: 192.168.10.0 and subnet mask: 255.255.255.240 then calculate total number of subnets and numbers of hosts per subnet.

[1+2+2]

Answer: Subnetting: Subnetting is the process of dividing a single physical network into multiple smaller logical networks (subnets). It improves network performance, enhances security, and allows efficient use of IP addresses. Subnetting is done by borrowing bits from the host portion of an IP address to create a subnet identifier. **Benefits of Subnetting:** • Reduces network congestion by limiting broadcast domains • Improves security by isolating network segments • Simplifies network management and troubleshooting • Allows flexible allocation of IP addresses • Enables logical organization by department or function **Given:** • Network Address: 192.168.10.0 • Subnet Mask: 255.255.255.240 **Step 1: Convert Subnet Mask to Binary** 255.255.255.240 = 11111111.11111111.11111111.11110000 **Step 2: Determine Number of Network Bits** • Number of network bits = 28 (28 ones in the mask) • This is a /28 network • Original Class C network has 24 network bits • Subnet bits borrowed = 28 - 24 = 4 bits **Step 3: Calculate Number of Subnets** Number of subnets = $2^{(\text{subnet bits})} = 2^4 = 16$ subnets **Step 4: Calculate Number of Host Bits** • Total bits = 32 • Network bits = 28 • Host bits = 32 - 28 = 4 bits **Step 5: Calculate Hosts per Subnet** Number of hosts per subnet = $2^{(\text{host bits})} - 2 = 2^4 - 2 = 16 - 2 = 14$ hosts per subnet (We subtract 2 because first address is network address and last is broadcast address) **Summary:** ParameterValue Network Address192.168.10.0 Subnet Mask255.255.255.240 (/28) Number of Subnets16 Hosts per Subnet14 usable hosts Block Size16 addresses per subnet First Subnet192.168.10.0 - 192.168.10.15 Second Subnet192.168.10.16 - 192.168.10.31 Last Subnet192.168.10.240 - 192.168.10.255 **Example Subnet Ranges:** • Subnet 1: Network=192.168.10.0, Usable=192.168.10.1 to .14, Broadcast=192.168.10.15 • Subnet 2: Network=192.168.10.16, Usable=192.168.10.17 to .30, Broadcast=192.168.10.31 • Subnet 16: Network=192.168.10.240, Usable=192.168.10.241 to .254, Broadcast=192.168.10.255

7. Draw a User Datagram format. Explain UDP operations.

[2+3]

Answer: UDP (User Datagram Protocol): UDP is a connectionless, unreliable transport layer protocol in the TCP/IP suite. It provides minimal overhead and fast transmission but does not guarantee delivery, ordering, or duplicate protection.

UDP Header Format (8 bytes total):

0	7	8	15	16	23	24	31
+-----+-----+-----+		Source Port		Destination Port		+-----+-----+-----+	
+-----+-----+-----+		Checksum		+-----+-----+-----+		Length	
+-----+-----+-----+		+-----+-----+-----+		+-----+-----+-----+		+-----+-----+-----+	

Fields:

- Source Port (16 bits):** Port number of the sender. Optional; set to 0 if not used.
- Destination Port (16 bits):** Port number of the receiver. Identifies the application process.
- Length (16 bits):** Total length of UDP segment (header + data) in bytes. Minimum value is 8 (header only).
- Checksum (16 bits):** Optional error-checking field. Covers header, data, and pseudo-header (IP addresses).

UDP Operations:

- Connectionless Communication:** - No connection establishment (no handshake like TCP) - Each datagram is independent - Sender simply formats the datagram and passes to IP - Fast but unreliable delivery
- Encapsulation:** - Application data is encapsulated with UDP header - UDP segment is passed to IP layer - IP adds its own header and routes the packet
- Demultiplexing:** - Receiver uses destination port number to deliver data to correct application - Multiple applications can use UDP simultaneously on different ports
- No Error Recovery:** - UDP checksum detects errors but does not correct them - Corrupted packets are silently discarded (no retransmission) - Applications must handle error recovery if needed
- No Flow Control:** - No mechanism to prevent sender from overwhelming receiver - No sliding window or acknowledgment system
- No Ordering:** - Packets may arrive out of order - No sequence numbers to reorder packets - Applications handle ordering if required

Applications Using UDP:

- DNS (Domain Name System) queries
- Streaming media (video/audio)
- Online gaming
- VoIP (Voice over IP)
- SNMP (Simple Network Management Protocol)
- TFTP (Trivial File Transfer Protocol)

Advantages: Low latency, minimal overhead, no connection setup delay, suitable for real-time applications

Disadvantages: Unreliable, no congestion control, no guaranteed delivery

8. Write short notes on (Any Two): a) DNS b) Public Key Cryptography c) VPN

[2.5+2.5]

Answer: a) DNS (Domain Name System): DNS is a hierarchical, distributed naming system that translates human-readable domain names (like www.google.com) into IP addresses (like 142.250.185.78). It acts as the "phonebook of the Internet." **How DNS Works:** 1. User types www.example.com in browser 2. Browser checks local cache, then OS cache 3. If not found, query goes to configured DNS resolver (usually ISP's DNS) 4. Resolver queries Root DNS servers → TLD servers (.com) → Authoritative DNS servers 5. Authoritative server returns the IP address 6. Resolver caches the result and returns to client 7. Browser connects to the IP address **DNS Record Types:** • A: Maps domain to IPv4 address • AAAA: Maps domain to IPv6 address • CNAME: Alias for another domain • MX: Mail exchange server • NS: Name server for the domain • TXT: Text records (SPF, DKIM) • PTR: Reverse DNS lookup **Benefits:** Human-friendly names, load distribution via multiple IPs, easy migration between servers --- **b) Public Key Cryptography:** Public key cryptography (asymmetric cryptography) uses a pair of keys: a public key for encryption and a private key for decryption. The public key can be freely shared, while the private key must be kept secret. **Key Characteristics:** • Two mathematically related but distinct keys • Computationally infeasible to derive private key from public key • Solves the key distribution problem of symmetric cryptography **How it works for Encryption:** 1. Sender encrypts message with recipient's public key 2. Only recipient can decrypt with their private key **How it works for Digital Signatures:** 1. Sender hashes the message and encrypts hash with their private key 2. Anyone can verify by decrypting with sender's public key and comparing hashes **Common Algorithms:** • RSA (Rivest-Shamir-Adleman): Based on factoring large primes • ECC (Elliptic Curve Cryptography): More efficient, smaller key sizes • Diffie-Hellman: Key exchange protocol **Applications:** SSL/TLS certificates, secure email (PGP), code signing, blockchain, VPN authentication **Advantages:** No need for secure key exchange, supports digital signatures, non-repudiation **Disadvantages:** Slower than symmetric cryptography, larger key sizes required --- **c) VPN (Virtual Private Network):** A VPN creates a secure, encrypted connection over a public network (like the Internet), allowing remote users to access a private network as if they were directly connected to it. **How VPN Works:** 1. User connects to VPN client and authenticates 2. VPN client establishes encrypted tunnel to VPN server 3. All traffic between client and server is encrypted 4. VPN server forwards requests to destination on behalf of client 5. Responses return through the encrypted tunnel **Types of VPN:** • **Remote Access VPN:** Individual users connect to corporate network (e.g., employees working from home) • **Site-to-Site VPN:** Connects entire networks at different locations (e.g., branch offices to headquarters) • **SSL VPN:** Uses web browser for remote access (no client installation) • **IPsec VPN:** Operates at network layer, provides strong security **VPN Protocols:** • OpenVPN: Open-source, highly secure, uses SSL/TLS • IPsec: Industry standard for site-to-site connections • L2TP/IPsec: Combines L2TP tunneling with IPsec encryption • WireGuard: Modern, fast, lightweight protocol **Benefits:** Privacy and anonymity, bypass geo-restrictions, secure remote work, protection on public Wi-Fi **Limitations:** Reduced speed due to encryption, potential logging by VPN providers, not completely anonymous

Group C

Attempt any TWO questions. [2×10=20]

9. Critically analyze the OSI reference model.

[10]

Answer: OSI Reference Model (Open Systems Interconnection): The OSI model is a conceptual framework developed by ISO in 1984 that standardizes network communication into seven distinct layers. Each layer has specific functions and communicates with adjacent layers through well-defined interfaces. **The Seven Layers:**

- 7. Application Layer:** • Closest to the end user • Provides network services to applications (HTTP, FTP, SMTP, DNS) • Handles user authentication, data formatting, and presentation • Example: Web browser, email client
- 6. Presentation Layer:** • Translates data between application and network formats • Handles encryption/decryption, compression/decompression • Data representation (ASCII, EBCDIC, JPEG, MPEG) • Example: SSL/TLS encryption, data compression
- 5. Session Layer:** • Establishes, manages, and terminates connections (sessions) • Handles dialogue control (full-duplex, half-duplex) • Synchronization using checkpoints for long transmissions • Example: NetBIOS, RPC, PPTP
- 4. Transport Layer:** • End-to-end communication and data delivery • Segmentation and reassembly of data • Flow control and error recovery • TCP (reliable, connection-oriented) and UDP (unreliable, connectionless)
- 3. Network Layer:** • Logical addressing (IP addressing) • Routing and path determination between networks • Packet forwarding and fragmentation • Protocols: IP, ICMP, ARP, RIP, OSPF, BGP
- 2. Data Link Layer:** • Physical addressing (MAC addresses) • Framing, error detection (CRC), and flow control • Divided into LLC (Logical Link Control) and MAC (Media Access Control) • Protocols: Ethernet, Wi-Fi (802.11), PPP, HDLC
- 1. Physical Layer:** • Transmits raw bits over physical medium • Defines electrical, mechanical, and procedural specifications • Cables, connectors, hubs, repeaters, signal encoding • Examples: Ethernet cables, fiber optics, radio waves

Critical Analysis - Advantages:

- 1. Modularity:** Changes in one layer don't affect others
- 2. Standardization:** Provides common framework for vendors
- 3. Interoperability:** Different systems can communicate
- 4. Troubleshooting:** Easy to isolate network problems by layer
- 5. Education:** Excellent teaching tool for understanding networking
- 6. Protocol Independence:** Allows multiple protocols at each layer

Critical Analysis - Disadvantages:

- 1. Complexity:** Seven layers can be unnecessarily complex
- 2. Performance Overhead:** Multiple layers add processing overhead
- 3. Not Widely Implemented:** TCP/IP (4 layers) became the de facto standard
- 4. Some Layers Redundant:** Presentation and Session layers often merged with Application layer in practice
- 5. Designed Before Internet:** Some concepts don't match modern networking reality
- 6. Strict Layering Inefficient:** Some protocols (like TCP/IP) violate strict layering for performance

Comparison with TCP/IP: AspectOSITCP/IP Layers74 DevelopmentModel first, then protocolsProtocols first, then model ApproachVertical (strict)Horizontal (loose) AdoptionTheoretical referencePractical implementation Protocol SupportGenericTCP/IP specific

Conclusion: While OSI is not directly implemented, its concepts and terminology are fundamental to networking education and troubleshooting. The layered approach remains valid, and OSI terminology (like "Layer 2 switch" or "Layer 3 router") is widely used in the industry.

10. Explain the random-access protocols under the multiple access taxonomy.

[10]

Answer: Random Access Protocols: Random access protocols allow stations to access the channel randomly without prior coordination. When multiple stations transmit simultaneously, collisions may occur. These protocols are also called contention protocols.

1. ALOHA: The simplest random access protocol developed at University of Hawaii.

Pure ALOHA:

- Stations transmit whenever they have data
- No synchronization or coordination
- If collision occurs, station waits random time and retransmits
- Vulnerable time = $2 \times$ frame transmission time
- Maximum throughput = 18.4% ($1/2e$)

Slotted ALOHA:

- Time is divided into discrete slots (equal to frame transmission time)
- Stations can only transmit at slot boundaries
- Reduces collisions by half
- Vulnerable time = $1 \times$ slot time
- Maximum throughput = 36.8% ($1/e$)

2. CSMA (Carrier Sense Multiple Access): Stations listen to the channel before transmitting.

a) 1-Persistent CSMA:

- Station senses channel before transmitting
- If idle: transmits immediately (probability 1)
- If busy: waits until channel becomes idle, then transmits immediately
- Problem: High collision probability when multiple stations waiting

b) Non-Persistent CSMA:

- If channel idle: transmits immediately
- If channel busy: waits random time before sensing again
- Reduces collisions but increases delay

c) p-Persistent CSMA:

- If channel idle: transmits with probability p , or defer to next slot with probability $(1-p)$
- If channel busy: waits until next slot and senses again
- Used in slotted channels, balances between 1-persistent and non-persistent

3. CSMA/CD (CSMA with Collision Detection): Used in Ethernet (IEEE 802.3).

Operation:

1. Station senses channel before transmitting
2. If idle, begins transmission while continuing to sense
3. If collision detected during transmission, station stops immediately
4. Sends jam signal to inform all stations of collision
5. Waits random time (binary exponential backoff) and retries

Binary Exponential Backoff:

- After first collision: wait 0 or 1 slot times
- After second collision: wait 0, 1, 2, or 3 slot times
- After n th collision: wait 0 to $2^n - 1$ slot times (capped at 1023)
- After 16 attempts, frame is discarded

Throughput: Much higher than ALOHA, approaches 100% under light load

Limitation: Requires minimum frame size to ensure collision detection; not suitable for wireless

4. CSMA/CA (CSMA with Collision Avoidance): Used in wireless networks (IEEE 802.11 Wi-Fi).

Why CSMA/CD doesn't work for Wireless:

- Hidden terminal problem: Two stations can't hear each other but both reach the access point
- Signal strength decay: Can't reliably detect collisions while transmitting
- Fading and interference make collision detection unreliable

Operation:

1. Station senses channel before transmitting
2. If idle: waits for DIFS (Distributed Inter-Frame Space), then transmits
3. If busy: waits until channel is idle, then waits additional DIFS
4. Uses RTS/CTS (Request to Send / Clear to Send) handshake:
 - Sender sends RTS to AP
 - AP responds with CTS
 - Other stations hearing CTS defer transmission
 - Sender transmits data
 - AP sends ACK after successful reception
5. Uses NAV (Network Allocation Vector) to track expected channel busy time

5. Controlled Access Protocols (for comparison):

- **Reservation:** Stations reserve slots before transmitting
- **Polling:** Master station invites slave stations to transmit
- **Token Passing:** Stations pass a token; only token holder can transmit

Comparison Table:

Protocol	Collision Handling	Throughput	Usage
Pure ALOHA	Retransmit after random delay	18%	Historical
Slotted ALOHA	Retransmit after random delay	37%	Historical
CSMA	Various strategies	Varies	Theoretical
CSMA/CD	Collision detection + backoff	~90%	Wired Ethernet
CSMA/CA	Collision avoidance (RTS/CTS)	~70%	Wireless (Wi-Fi)

11. Explain the IPv4 Header format in detail.

[10]

Answer: IPv4 Header Format: The IPv4 header is variable in size (minimum 20 bytes, maximum 60 bytes) and contains all necessary information for routing and delivering packets across networks.

0	1	2	3	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1				
+-+										[Version]		[IHL]		[Type of Service]		[Total Length]													
+-+										[Identification]		[Flags]		[Fragment Offset]															
+-+										[Time to Live]		[Protocol]		[Header Checksum]															
+-+										[Source IP Address]										[Destination IP Address]									
+-+										[Options (Variable, 0-40 bytes)]																			

Field Descriptions:

- Version (4 bits):** - Indicates IP version (4 for IPv4) - Allows routers to process packet correctly
- IHL - Internet Header Length (4 bits):** - Length of header in 32-bit words - Minimum value = 5 ($5 \times 4 = 20$ bytes) - Maximum value = 15 ($15 \times 4 = 60$ bytes)
- Type of Service / DSCP (8 bits):** - Specifies quality of service parameters - Originally: Precedence, Delay, Throughput, Reliability bits - Now: Differentiated Services Code Point (DSCP) for QoS - ECN (Explicit Congestion Notification) bits for congestion signaling
- Total Length (16 bits):** - Entire packet size (header + data) in bytes - Maximum value = 65,535 bytes - Data length = Total Length - (IHL $\times$ 4)
- Identification (16 bits):** - Unique identifier for each packet - Used for reassembling fragmented packets - All fragments of same packet have same identification number
- Flags (3 bits):** - Bit 0: Reserved (must be 0) - Bit 1 (DF - Don't Fragment): If set, packet cannot be fragmented - Bit 2 (MF - More Fragments): If set, more fragments follow; 0 for last fragment
- Fragment Offset (13 bits):** - Position of fragment relative to original packet data - Measured in units of 8 bytes - First fragment has offset 0
- Time to Live / TTL (8 bits):** - Maximum number of hops (routers) packet can traverse - Decrement by 1 at each router - Packet discarded when TTL reaches 0 (prevents infinite loops) - Typical initial values: 64 (Linux), 128 (Windows), 255 (routers)
- Protocol (8 bits):** - Identifies transport layer protocol - 1 = ICMP, 6 = TCP, 17 = UDP, 41 = IPv6, 50 = ESP, 89 = OSPF
- Header Checksum (16 bits):** - Error detection for header only (not data) - Recalculated at each router (TTL changes) - If checksum fails, packet is discarded
- Source IP Address (32 bits):** - IPv4 address of sender
- Destination IP Address (32 bits):** - IPv4 address of intended receiver
- Options (Variable, 0-40 bytes):** - Rarely used in practice - Security, strict source routing, loose source routing, record route, timestamp - Must be padded to 32-bit boundary

Key Points:

- Minimum header size = 20 bytes (when no options)
- Header size must be multiple of 4 bytes
- Options field allows extension but reduces efficiency
- Checksum covers only header, not payload (higher layers handle data integrity)
- Fragmentation allows packets to traverse networks with smaller MTU

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.